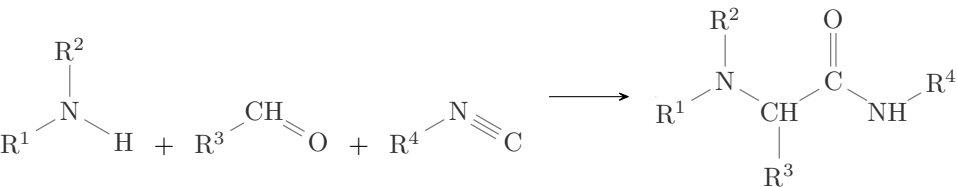


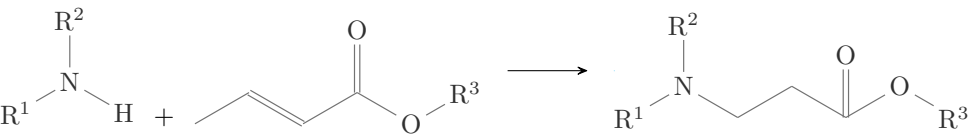
Ugi three-component reaction

amine head, aldehyde and isocyanide combine in one step; no carboxylic-acid reactant component



Repeated aza-Michael addition

the head N-H adds across the acrylate double bond; the ester is untouched



Repeated reductive amination

one new carbon-nitrogen bond; the carbonyl oxygen leaves as water

